

Q1. Create a directory named mid_exam with the following subdirectories:

- documents
- scripts
- logs
- backup

Ans:

```
mkdir -p mid_exam/{documents,scripts,logs,backup}
```

```
(kali㉿kali)-[~]
└─$ mkdir -p mid_exam/{documents,scripts,logs,backup}

(kali㉿kali)-[~]
└─$ ls mid_exam
backup documents logs scripts

(kali㉿kali)-[~]
└─$
```

Q2. Create a file named employee_info.txt inside the documents directory and add company information to it. Then copy the file to the backup directory and rename it as employee_backup.txt.

Ans:

1. cd mid_exam/documents
2. cat >> employee_info.txt
3. cp employee_info.txt ../backup/employee_backup.txt

```
(kali㉿kali)-[~]
└─$ cd mid_exam/documents

(kali㉿kali)-[~/mid_exam/documents]
└─$ cat >> employee_info.txt
Name: Safayet Ullah
ID: 01

Name: Ullah Safayet
ID: 02

Company Name: Southeast University
Location: Tejgaon, Dhaka
VC: Safayet Ullah
```

```
(kali㉿kali)-[~/mid_exam/documents]
└─$ cp employee_info.txt ../backup/employee_backup.txt

(kali㉿kali)-[~/mid_exam/documents]
└─$ cd ../backup

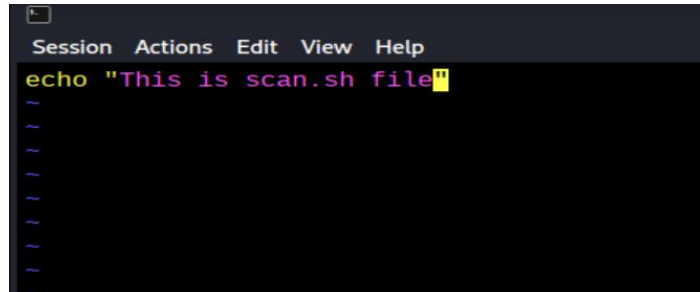
(kali㉿kali)-[~/mid_exam/backup]
└─$ ll
total 4
-rw-rw-r-- 1 kali kali 136 May 12 08:20 employee_backup.txt

(kali㉿kali)-[~/mid_exam/backup]
└─$
```

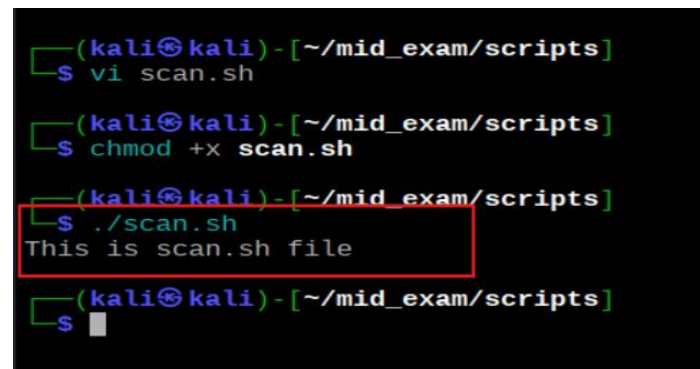
Q3. Create a shell script named scan.sh inside the scripts directory containing a simple echo command. Give execute permission only to the owner and run the script.

Ans:

1. **Create Shell:** vi scan.sh
2. Go to insert mode [Press 'i']
3. Save and exit [:wq]
4. **Give Permission:** chmod +x scan.sh
5. **Run:** ./scan.sh

A screenshot of a text editor window with a dark background. The menu bar at the top includes 'Session', 'Actions', 'Edit', 'View', and 'Help'. The main text area contains the command 'echo "This is scan.sh file"' followed by a cursor. Below the command, there are several empty lines, each starting with a tilde (~) character.

```
Session Actions Edit View Help
echo "This is scan.sh file"
~
~
~
~
~
~
```

A screenshot of a terminal window with a dark background. The prompt is '(kali㉿kali)-[~/mid_exam/scripts]'. The user enters 'vi scan.sh', then 'chmod +x scan.sh', and finally './scan.sh'. The output of the script, 'This is scan.sh file', is displayed on the next line. A red rectangle highlights the command './scan.sh' and its output.

```
(kali㉿kali)-[~/mid_exam/scripts]
$ vi scan.sh

(kali㉿kali)-[~/mid_exam/scripts]
$ chmod +x scan.sh

(kali㉿kali)-[~/mid_exam/scripts]
$ ./scan.sh
This is scan.sh file

(kali㉿kali)-[~/mid_exam/scripts]
$
```

Q4. Collect the following system information:

IP address
Hostname
Logged-in user
Ping test results

Save all outputs in a file named network_report.txt inside the logs directory.

Ans:

1. **Create File:** touch network_report.txt
2. echo "IP Address:" >> network_report.txt
hostname -I >> network_report.txt
3. echo "Hostname:" >> network_report.txt
hostname >> network_report.txt
4. echo "Logged-in User:" >> network_report.txt
whoami >> network_report.txt
5. echo "Ping Test Results:" >> network_report.txt
ping 8.8.8.8 >> network_report.txt

```
(kali㉿kali)-[~/mid_exam/logs]
$ cat network_report.txt
IP Address:
192.168.220.128 172.17.0.1
Hostname:
kali
Logged-in User:
kali
Ping Test Results:
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2041ms
```

Q5: Run a ping test on 8.8.8.8 and perform a basic Nmap scan on scanme.nmap.org. Save the results in a file named scan_result.txt inside the logs directory.

Ans:

1. ping 8.8.8.8 >>scan_result.txt
2. nmap -sC scanme.nmap.org >> scan_result.txt

```
(kali@kali)-[~/mid_exam/logs]
$ cat scan_result.txt
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=36.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=53.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=71.7 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 35.957/53.838/71.715/14.598 ms
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-12 08:49 -0400
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.22 seconds
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-12 08:50 -0400
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-12 08:50 -0400
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.31 seconds
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-12 08:51 -0400
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-12 08:53 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00060s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 10.37 seconds
```

Q6: Display currently running processes and monitor system activity using terminal-based utilities.

Ans:

1. Running Process: ps
2. System Activity: top

```
(kali@kali)-[~/mid_exam/logs]
$ top
top - 08:58:44 up 2:36, 1 user, load average: 0.08, 0.08, 0.07
Tasks: 257 total, 1 running, 256 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.7 us, 1.0 sy, 0.0 ni, 98.2 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem: 7907.5 total, 242.0 free, 2138.0 used, 5840.2 buff/cache
MiB Swap: 953.7 total, 953.7 free, 0.0 used, 5769.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
 1087 root        20   0 456644 158256 88172 S   1.7   2.0   1:42.08 Xorg
 376 root        20   0 50476 18364 16892 S   0.3   0.2   0:01.32 systemd-journal
 586 root        20   0 253024 10344 8584 S   0.3   0.1   0:25.56 vmtoolsd
 716 root        20   0 336832 20212 17120 S   0.3   0.2   0:01.15 NetworkManager
1902 kali        20   0 1221888 139916 91332 S   0.3   1.7   0:44.78 xfwm4
2031 kali        20   0 315124 64472 22040 S   0.3   0.8   0:33.98 wrapper-2.0
2441 kali        20   0 918184 48780 36140 S   0.3   0.6   0:26.63 vmtoolsd
2792 kali        20   0 802088 62528 52724 S   0.3   0.8   0:13.36 qterminal
2941 kali        20   0 3022536 419752 181992 S   0.3   5.2   1:21.99 firefox-esr
3343 kali        20   0 2443196 81640 67580 S   0.3   1.0   0:08.97 Web Content
3416 kali        20   0 2437644 81168 67184 S   0.3   1.0   0:08.80 Web Content
   1 root        20   0 24624 15380 11516 S   0.0   0.2   0:02.90 systemd
   2 root        20   0      0      0      0 S   0.0   0.0   0:00.03 kthreadd
   3 root        20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-syncg_wg
```

Q7: Generate both MD5 and SHA256 hashes of the employee_info.txt file and save the outputs in a file named hash_report.txt inside the logs directory.

Ans:

1. `md5sum employee_info.txt >> hash_report.txt`
2. `sha256sum employee_info.txt >> hash_report.txt`

```
(kali㉿kali)-[~/mid_exam/documents]
$ md5sum employee_info.txt >> hash_report.txt

(kali㉿kali)-[~/mid_exam/documents]
$ sha256sum employee_info.txt >> hash_report.txt

(kali㉿kali)-[~/mid_exam/documents]
$ cat hash_report.txt
672a6117a52ae8397cc4f7a49098b477 employee_info.txt
92c0c1e54a42c5b57a6884bb1c68d2f83af2c06349dcd94f8ecc72d769749cd7 employee_info.txt

(kali㉿kali)-[~/mid_exam/documents]
$
```

Q8: Investigate authentication logs by displaying the last 15 lines and searching for the keyword "failed".

Ans:

1. `tail -n 15 /var/log/auth.log`
2. `grep "failed" /var/log/auth.log`

```
(kali㉿kali)-[~/mid_exam/documents]
$ grep "failed" /var/log/auth.log
grep: /var/log/auth.log: No such file or directory

(kali㉿kali)-[~/mid_exam/documents]
$
```

Check : `ls /var/log`

```
(kali㉿kali)-[~/mid_exam/documents]
$ ls /var/log/
alternatives.log      bttmp.1              mosquitto            syslog               vmware-vmtoolsd-root.log
alternatives.log.1    dpkg.log             mysql                vmware-network.1.log vmware-vmtoolsd-kali.log
apache2               dpkg.log.1           nginx               vmware-network.2.log vmware-vmtoolsd-root.log
apt                   fontconfig.log       notus-scanner        vmware-network.3.log vmware-vmusr-kali.log
boot.log              gvm                  openvpn              vmware-network.4.log wtmp
boot.log.1            inetd.conf            private              vmware-network.5.log wttmp.db
boot.log.2            journal              README               vmware-network.6.log Xorg.0.log
boot.log.3            lastlog               redis                vmware-network.7.log Xorg.0.log.old
boot.log.4            lightdm               runit                vmware-network.8.log Xorg.1.log
boot.log.5            macchanger.log        samba                vmware-network.9.log
boot.log.6            macchanger.log.1.gz  speech-dispatcher    vmware-network.log   vmware-vmtoolsd-kali.log
boot.log.7            macchanger.log.2.gz  stunnel4             vmware-vmtoolsd-root.log
bttmp                 macchanger.log.3.gz
```


Q8: Create a bash script named `system_info.sh` that displays:

Current date
Current user
Current working directory
Hostname

Make the script executable and run it.

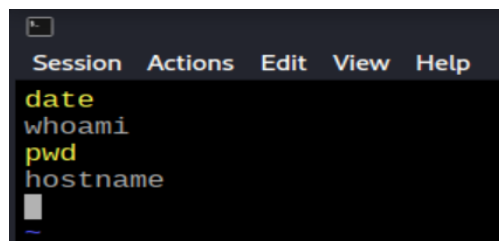
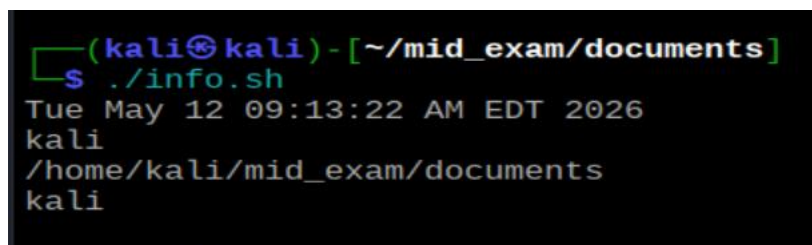
Ans:

1. `vi info.sh`

- `date`
- `whoami`
- `pwd`
- `hostname`

2. `chmod +x info.sh`

3. `./info.sh`

A screenshot of a terminal window with a dark background. The window has a menu bar at the top with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal content shows the script's output: 'date', 'whoami', 'pwd', and 'hostname', each on a new line. A cursor is visible at the bottom.A screenshot of a terminal window showing the execution of the script. The prompt is '(kali@kali)-[~/mid_exam/documents]'. The command './info.sh' has been entered. The output displayed is: 'Tue May 12 09:13:22 AM EDT 2026', 'kali', '/home/kali/mid_exam/documents', and 'kali'.